1. What is a UAV vs a Drone?

UAV (Unmanned Aerial Vehicle) is the formal engineering and industry term for any aircraft that flies without a human pilot physically on board. It covers the whole system: the airframe, flight controller, sensors, communication link, and ground control software. This is the term used in defence, aviation regulation, and engineering documentation.

Drone is the everyday, popular term for the same category of aircraft. In casual use it is often pictured as a small multirotor (quadcopter), but technically a drone and a UAV describe the same thing. In short: every drone is a UAV; “UAV” is just the precise, professional name used in technical and industry contexts.

2. Types of Drones

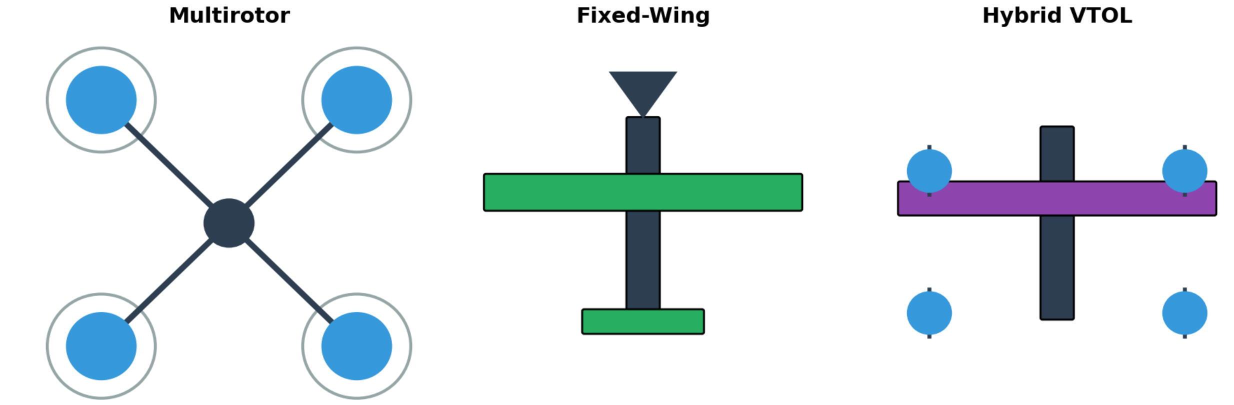

Figure 1: The three major UAV airframe categories.

Type	How it flies	Strength	Limitation
Multirotor	Multiple rotors generate lift directly (vertical thrust)	Hovers, vertical take-off/landing, very maneuverable	Short flight time, high power consumption
Fixed-wing	A single wing generates lift as air flows over it in forward flight	Long endurance and range, fuel/battery efficient	Cannot hover, needs runway or launcher
Hybrid VTOL	Combines rotors for vertical lift with a wing for forward flight	Vertical take-off + long-range cruise efficiency	Mechanically complex, heavier, costlier

3. Basic UAV Components

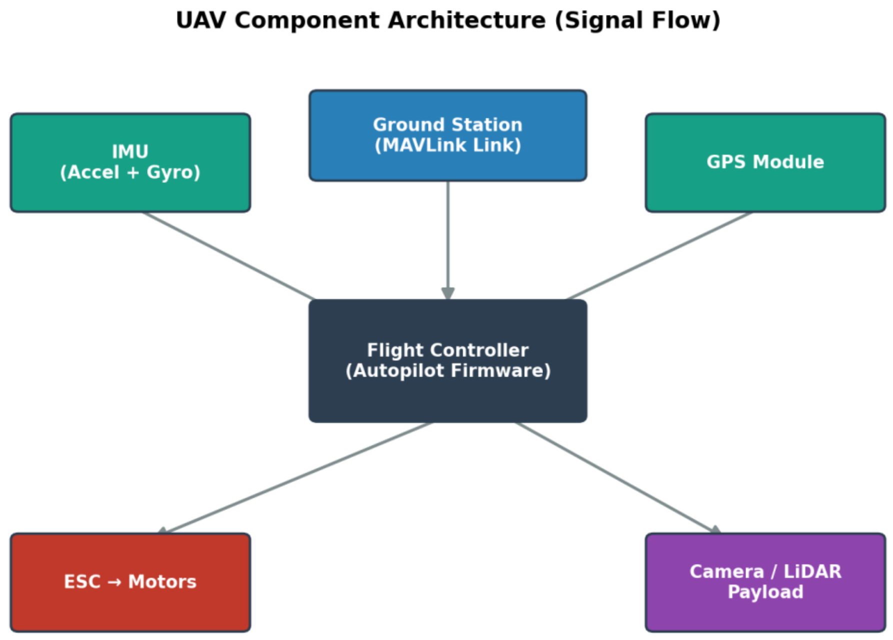

Figure 2: How the core components exchange data with the flight controller.

- **Flight Controller (FC):** The “brain” of the UAV. An onboard computer that runs the autopilot firmware, fuses sensor data, and sends motor commands to keep the craft stable and flying the intended path.
- **ESC (Electronic Speed Controller):** Sits between the flight controller and each motor. It converts the FC's digital command into the precise electrical power needed to spin a motor at the requested speed.
- **GPS Module:** Provides global position (latitude/longitude), altitude, and ground speed. Essential for autonomous waypoint navigation and Return-To-Launch (RTL).
- **IMU – Accelerometer + Gyroscope:** Measures linear acceleration and rotational rate on all three axes. This tells the FC how the drone is tilting or accelerating many times per second, which is the core input for stabilization.
- **LiDAR / Camera Payload:** The mission-specific sensor carried by the UAV – e.g. a camera for aerial photography/mapping or a LiDAR scanner for 3D terrain and obstacle data.

4. Key Concepts: PID Control, MAV Link, Autopilot Firmware

4.1 PID Control

PID stands for **Proportional – Integral – Derivative**. It is the feedback control algorithm the flight controller uses to keep the drone at a target value (e.g. a target altitude or attitude angle).

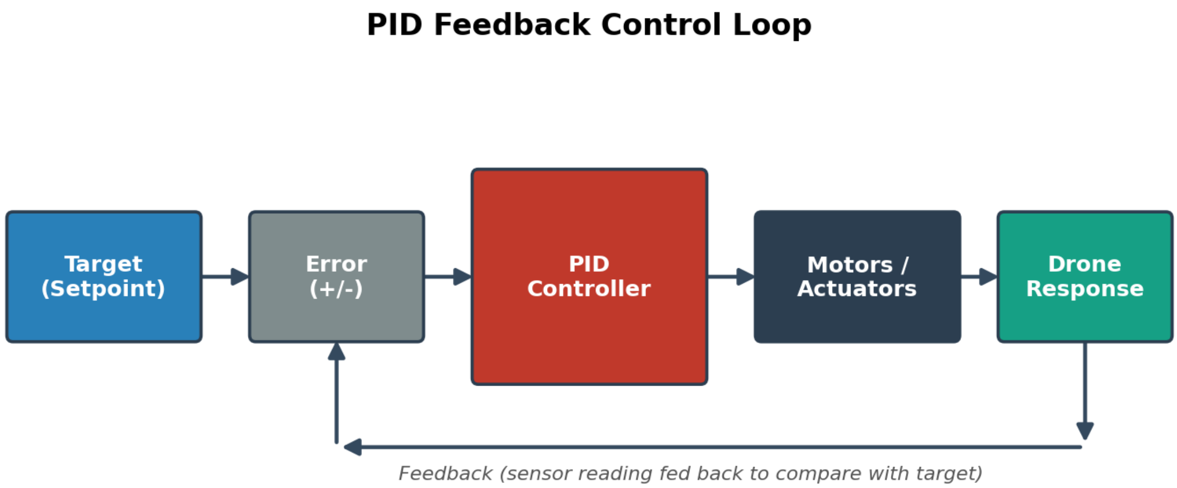

Figure 3: The PID feedback loop – the controller continuously compares the current state to the target and adjusts the motors.

- **P (Proportional):** reacts to the present error – bigger error, bigger correction.
- **I (Integral):** reacts to accumulated past error, removing any lingering offset.
- **D (Derivative):** reacts to how fast the error is changing, which damps overshoot.

4.2 MAV Link

MAV Link (Micro Air Vehicle Link) is a lightweight, standardized messaging protocol used to send telemetry and commands between a drone's flight controller, a ground control station, and companion computers/scripts (such as the Drone Kit script in Part 3). It defines a common “language” so the heartbeat, GPS, battery, and command messages mean the same thing across Auto Pilot, PX4, Mission Planner, Ground Control, and custom Python code.

4.3 Autopilot Firmware

Autopilot firmware is software that actually runs on the flight controller hardware. It implements sensor fusion, the PID control loops, mission/waypoint logic, failsafe (e.g. low battery or lost signal behavior), and the MAV Link interface. **AUTO Pilot** and **PX4** are the two leading open-source autopilot firmware projects used across the drone indus